



8063 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	HR 8799 bcd roll 1	NIRSpec IFU Spectroscopy	(1) HR_8799
	2	HR 8799 bcd roll 2	NIRSpec IFU Spectroscopy	(1) HR_8799
	3	HIP 65426 b 1-2um roll 1	NIRSpec IFU Spectroscopy	(3) HIP-65426
	4	HIP 65426 b 1-2um roll 2	NIRSpec IFU Spectroscopy	(3) HIP-65426
	5	HIP 65426 b 3-5um roll 1	NIRSpec IFU Spectroscopy	(3) HIP-65426

Folder	Observation	Label	Observing Template	Science Target
	6	HIP 65426 b 3-5um roll 2	NIRSpec IFU Spectroscopy	(3) HIP-65426
	7	bet Pic b roll 1	NIRSpec IFU Spectroscopy	(4) -bet-Pic
	8	bet Pic b roll 2	NIRSpec IFU Spectroscopy	(4) -bet-Pic
	9	YSES 1 b roll 1	NIRSpec IFU Spectroscopy	(5) TYC-8998-760-1-b
	10	YSES 1 b roll 2	NIRSpec IFU Spectroscopy	(5) TYC-8998-760-1-b
	11	kap And b roll 1	NIRSpec IFU Spectroscopy	(6) -kap-And
	12	kap And b roll 2	NIRSpec IFU Spectroscopy	(6) -kap-And
	13	HR 2562 b roll 1	NIRSpec IFU Spectroscopy	(7) HR-2562
	14	HR 2562 b roll 2	NIRSpec IFU Spectroscopy	(7) HR-2562

ABSTRACT

We propose to measure the abundances of four refractories (Potassium, K; Iron, Fe; Sodium, Na; Sulfur, S) in eight massive, directly imaged gas giant exoplanets (5-15 MJup; 10-160 au) with JWST/NIRSpec IFU. Refractories are elements that reside exclusively in solid form in the protoplanetary disk. In contrast, volatiles (e.g., C, O) evaporate from icy pebbles as they continuously migrate inward and cross evaporation fronts due to drag in the planet-forming disk. By measuring the relative enrichment of individual volatiles and refractories, the goal of this program is to yield new insight into the enigmatic formation mechanisms of gas giant planets on wide orbits and enable a comparison with the population of transiting planets. While young planets continuously accrete the volatile-rich gas, they can accrete various amounts of pebbles and planetesimals, but the refractory-rich pebbles get trapped by pressure bumps in the disk while larger planetesimals do not. Therefore, the volatile-to-refractory ratio (vol/ref) could indicate if massive directly-imaged planets formed preferentially through the accretion of pebbles (super-stellar vol/ref) or planetesimals (stellar vol/ref). However, disentangling the complex physical processes involved in planet-forming disks requires the measurement of multiple refractories and volatile species beyond even the recently measured and tantalizing C/S ratios. JWST now provides a unique opportunity to answer these fundamental questions by leveraging the unprecedented high-contrast sensitivity of the NIRSpec IFU at moderate spectral resolution ($R \sim 2,700$; 1-2 μm and 3-5 μm), which has not been achievable from the best facilities on the ground.

OBSERVING DESCRIPTION

We propose to use the high-resolution spectroscopy mode ($R \sim 2,700$) of the NIRSpec IFU to observe 8 planets around 6 separate stars (HR 8799 bcd, HIP 65426 b, HR 2562 b, bet Pic b, kap And b, YSES 1 b) in two filters: K, FeH, and Na can be detected between 1-2 μm (grism/filter: G140H/F100LP), and H₂S between 3-5 μm (G395H/F290LP). Other than YSES 1 b, the planets are separated by 0.5-2" from their bright host star ($J=3.5-7$) with a typical flux ratio around $1e-5$.

JWST Proposal 8063 (Created: Monday, July 14, 2025, 6:00:13PM Eastern Standard Time) - Overview

These bright stars will all saturate the PSF core, from 0.1" saturation to 0.5" for the brightest star beta Pictoris.

No target acquisition is required with the IFU.

We split each visit in two observatory rolls (~10 degrees apart) to avoid a chance alignment of the planet with bright speckles.

We set a position angle (PA) constraint to place the planet away from the saturate IFU slices on the detector and away from the diffraction spikes.

The number of groups and detector readout pattern is defined as a function of the stellar brightness to best manage saturation. For bet Pic and kap

And, the brightest stars, we use NRSRAPID with 2 groups to limit saturation as much as possible and provide the smallest inner working angle to

avoid the risk of saturating the planets.

A 16 small cycling dithers is chosen to mitigate NIRSpec spatial undersampling systematics. Only YSES 1 b, the easiest target, is set to 9 dithers.

We apply specific offsets for HR 8799 bcd and YSES 1 b to place the planets in the field of view.

Proposal 8063 - Targets - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

(7)	HR-2562	RA: 06 50 1.0151 (102.5042296d)	Proper Motion RA: 4.83 mas/yr
		Dec: -60 14 56.92 (-60.24914d)	Proper Motion Dec: 108.527 mas/yr
		Equinox: J2000	Parallax: 0.0294738"
			Epoch of Position: 2000

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.

Category=Star

Description=[F stars]

Proposal 8063 - Observation 1 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

Observation	Proposal 8063, Observation 1: HR 8799 bcd roll 1											Mon Jul 14 23:00:13 GMT 2025
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 1:1) Warning (Form): Data Excess over lower threshold											
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(HR 8799 bcd roll 1 (Obs 1)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
	(Visit 1:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	HR_8799	RA: 23 07 28.8754 (346.8703142d) Dec: +21 08 2.61 (21.13406d) Equinox: J2000				Proper Motion RA: 108.28365 mas/yr Proper Motion Dec: -50.03999999644293 mas/yr Parallax: 0.024462" Epoch of Position: 2016					
	Comments: This object was generated based on Gaia DR3 coordinates from the Gaia archive, hence the 2016 epoch.											
	It is not centered on the star, but the half-way point between HR 8799 b and HR 8799 d.											
	Category=Star											
	Description=[F stars]											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		16				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G140H/F100LP	NRSIRS2RAPID	8	10	false	true	NONE	16	160	21008.002	
Special Requirements	Aperture PA Range 10.97164917 to 18.97164917 Degrees (V3 232.0 to 240.0)											
	Group Observations 1, 2 within 30 Days Aperture PA Offset 2 from 1 by 10 to 16 Degrees (Same offsets in V3)											

Proposal 8063 - Observation 2 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

Observation	Proposal 8063, Observation 2: HR 8799 bcd roll 2											Mon Jul 14 23:00:13 GMT 2025
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 2:1) Warning (Form): Data Excess over lower threshold											
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(HR 8799 bcd roll 2 (Obs 2)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
	(Visit 2:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	HR_8799	RA: 23 07 28.8754 (346.8703142d)				Proper Motion RA: 108.28365 mas/yr			Comments: This object was generated based on Gaia DR3 coordinates from the Gaia archive, hence the 2016 epoch. It is not centered on the star, but the half-way point between HR 8799 b and HR 8799 d. Category=Star Description=[F stars] Extended=NO		
	Dec: +21 08 2.61 (21.13406d)				Proper Motion Dec: -50.03999999644293 mas/yr							
	Equinox: J2000				Parallax: 0.024462"							
					Epoch of Position: 2016							
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		16				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G140H/F100LP	NRSIRS2RAPID	8	10	false	true	NONE	16	160	21008.002	
Special Requirements	Aperture PA Range 10.97164917 to 28.97164917 Degrees (V3 232.0 to 250.0)											
	Group Observations 1, 2 within 30 Days Aperture PA Offset 2 from 1 by 10 to 16 Degrees (Same offsets in V3)											

Proposal 8063 - Observation 3 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

Observation	Proposal 8063, Observation 3: HIP 65426 b 1-2um roll 1											Mon Jul 14 23:00:13 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 3:1) Warning (Form): Data Excess over lower threshold											
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(HIP 65426 b 1-2um roll 1 (Obs 3)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(3)	HIP-65426	RA: 13 24 36.0977 (201.1504071d)				Proper Motion RA: -33.921 mas/yr					
	Dec: -51 30 16.04 (-51.50446d)				Proper Motion Dec: -18.917000011242635 mas/yr							
	Equinox: J2000				Parallax: 0.0093031"							
					Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
Category=Star												
Description=[A stars]												
Extended=NO												
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		16				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G140H/F100LP	NRSIRS2RAPID	8	9	false	true	NONE	16	144	18907.201	

Proposal 8063 - Observation 3 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

Special Requirements

Aperture PA Range 238.97164917 to 258.97164917 Degrees (V3 100.0 to 120.0)

Group Observations 3, 4 within 30 Days

Aperture PA Offset 4 from 3 by 8 to 20 Degrees (Same offsets in V3)

Proposal 8063 - Observation 4 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

Observation	Proposal 8063, Observation 4: HIP 65426 b 1-2um roll 2												Mon Jul 14 23:00:13 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 4:1) Warning (Form): Data Excess over lower threshold												
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(HIP 65426 b 1-2um roll 2 (Obs 4)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(3)	HIP-65426	RA: 13 24 36.0977 (201.1504071d)				Proper Motion RA: -33.921 mas/yr						
			Dec: -51 30 16.04 (-51.50446d)				Proper Motion Dec: -18.917000011242635 mas/yr						
			Equinox: J2000				Parallax: 0.0093031"						
							Epoch of Position: 2000						
			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
Template	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star												
	Description=[A stars]												
Dithers	Extended=NO												
	TA Method						HFF Readout Mode						
Spectral Elements	NONE						false						
	#	Dither Type		Size		Starting Point		Number of Points		Points			
	1	CYCLING		SMALL		1		16					
	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	G140H/F100LP	NRSIRS2RAPID	8	9	false	true	NONE	16	144	18907.201		

Proposal 8063 - Observation 4 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

Special Requirements

Aperture PA Range 238.97164917 to 258.97164917 Degrees (V3 100.0 to 120.0)

Group Observations 3, 4 within 30 Days

Aperture PA Offset 4 from 3 by 8 to 20 Degrees (Same offsets in V3)

Proposal 8063 - Observation 5 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

Observation	Proposal 8063, Observation 5: HIP 65426 b 3-5um roll 1											Mon Jul 14 23:00:13 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(HIP 65426 b 3-5um roll 1 (Obs 5)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(3)	HIP-65426	RA: 13 24 36.0977 (201.1504071d) Dec: -51 30 16.04 (-51.50446d) Equinox: J2000				Proper Motion RA: -33.921 mas/yr Proper Motion Dec: -18.917000011242635 mas/yr Parallax: 0.0093031" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[A stars] Extended=NO											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		16				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSIRS2RAPID	8	2	false	true	NONE	16	32	4201.6	

Proposal 8063 - Observation 5 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

Special Requirements

Aperture PA Range 238.97164917 to 258.97164917 Degrees (V3 100.0 to 120.0)

Group Observations 5, 6, Non-interruptible

Aperture PA Offset 6 from 5 by 8 to 20 Degrees (Same offsets in V3)

Proposal 8063 - Observation 6 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

Observation	Proposal 8063, Observation 6: HIP 65426 b 3-5um roll 2											Mon Jul 14 23:00:13 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(HIP 65426 b 3-5um roll 2 (Obs 6)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(3)	HIP-65426	RA: 13 24 36.0977 (201.1504071d)				Proper Motion RA: -33.921 mas/yr					
			Dec: -51 30 16.04 (-51.50446d)				Proper Motion Dec: -18.917000011242635 mas/yr					
			Equinox: J2000				Parallax: 0.0093031"					
							Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star											
	Description=[A stars]											
	Extended=NO											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		16				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSIRS2RAPID	8	2	false	true	NONE	16	32	4201.6	

Proposal 8063 - Observation 6 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

Special Requirements

Aperture PA Range 238.97164917 to 258.97164917 Degrees (V3 100.0 to 120.0)

Group Observations 5, 6, Non-interruptible

Aperture PA Offset 6 from 5 by 8 to 20 Degrees (Same offsets in V3)

Proposal 8063 - Observation 7 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

Observation	Proposal 8063, Observation 7: bet Pic b roll 1												Mon Jul 14 23:00:13 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(bet Pic b roll 1 (Obs 7)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(4)	-bet-Pic	RA: 05 47 17.0877 (86.8211988d)				Proper Motion RA: 5.16 mas/yr						
			Dec: -51 03 59.44 (-51.06651d)				Proper Motion Dec: 84.041 mas/yr						
			Equinox: J2000				Parallax: 0.0509307"						
							Epoch of Position: 2000						
			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
Template	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star												
	Description=[A stars]												
	Extended=NO												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points		
	1	CYCLING			SMALL		1		16				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	G140H/F100LP	NRSRAPID	2	7	false	true	NONE	16	112	3607.555		
	2	G395H/F290LP	NRSRAPID	4	3	false	true	NONE	16	48	2576.825		

Proposal 8063 - Observation 7 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?	
Special Requirements	Aperture PA Range 112.97164917 to 131.97164917 Degrees (V3 334.0 to 353.0) Group Observations 7, 8, Non-interruptible Aperture PA Offset 8 from 7 by 10 to 20 Degrees (Same offsets in V3)

Proposal 8063 - Observation 8 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

Observation	Proposal 8063, Observation 8: bet Pic b roll 2											Mon Jul 14 23:00:13 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(bet Pic b roll 2 (Obs 8)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(4)	-bet-Pic	RA: 05 47 17.0877 (86.8211988d)				Proper Motion RA: 5.16 mas/yr					
			Dec: -51 03 59.44 (-51.06651d)				Proper Motion Dec: 84.041 mas/yr					
			Equinox: J2000				Parallax: 0.0509307"					
							Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
Category=Star												
Description=[A stars]												
Extended=NO												
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		16				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G140H/F100LP	NRSRAPID	2	7	false	true	NONE	16	112	3607.555	
	2	G395H/F290LP	NRSRAPID	4	3	false	true	NONE	16	48	2576.825	

Proposal 8063 - Observation 8 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

Special Requirements

Aperture PA Range 112.97164917 to 131.97164917 Degrees (V3 334.0 to 353.0)

Group Observations 7, 8, Non-interruptible

Aperture PA Offset 8 from 7 by 10 to 20 Degrees (Same offsets in V3)

Proposal 8063 - Observation 9 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

Observation	Proposal 8063, Observation 9: YSES 1 b roll 1											Mon Jul 14 23:00:13 GMT 2025
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(YSES 1 b roll 1 (Obs 9)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(5)	TYC-8998-760-1-b	RA: 13 25 11.7773 (201.2990721d)				Proper Motion RA: -40.995708 mas/yr			Comments: These coordinates are the midpoint location between both planets Tyc 8998-760-1 b and c. By targeting this location with the NIRSpec IFU, both planets will be within the NIRSpec field of view. Category=Star Description=[Exoplanet Systems] Extended=NO		
	Dec: -64 56 22.97 (-64.93971d)				Proper Motion Dec: -17.733999970914738 mas/yr							
	Equinox: J2000				Parallax: 0.0106124"							
					Epoch of Position: 2016.0							
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G140H/F100LP	NRSIRS2RAPID	8	1	false	true	NONE	9	9	1181.7	
Special Requirements	Group Observations 9, 10, Non-interruptible Aperture PA Offset 10 from 9 by 10 to 20 Degrees (Same offsets in V3)											

Proposal 8063 - Observation 10 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

Observation	Proposal 8063, Observation 10: YSES 1 b roll 2											Mon Jul 14 23:00:13 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(YSES 1 b roll 2 (Obs 10)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(5)	TYC-8998-760-1-b	RA: 13 25 11.7773 (201.2990721d)				Proper Motion RA: -40.995708 mas/yr					
			Dec: -64 56 22.97 (-64.93971d)				Proper Motion Dec: -17.733999970914738 mas/yr					
			Equinox: J2000				Parallax: 0.0106124"					
							Epoch of Position: 2016.0					
	Comments: These coordinates are the midpoint location between both planets Tyc 8998-760-1 b and c. By targeting this location with the NIRSpec IFU, both planets will be within the NIRSpec field of view.											
	Category=Star											
	Description=[Exoplanet Systems]											
	Extended=NO											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G140H/F100LP	NRSIRS2RAPID	8	1	false	true	NONE	9	9	1181.7	
Special Requirements	Group Observations 9, 10, Non-interruptible											
	Aperture PA Offset 10 from 9 by 10 to 20 Degrees (Same offsets in V3)											

Proposal 8063 - Observation 11 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

Observation	Proposal 8063, Observation 11: kap And b roll 1											Mon Jul 14 23:00:13 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(kap And b roll 1 (Obs 11)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
	(Visit 11:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(6)	-kap-And	RA: 23 40 24.5076 (355.1021150d) Dec: +44 20 2.16 (44.33393d) Equinox: J2000				Proper Motion RA: 80.73 mas/yr Proper Motion Dec: -18.69999998689309 mas/yr Parallax: 0.019370000000000002" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star											
	Description=[B stars]											
Template	Extended=NO											
	TA Method						HFF Readout Mode					
NONEfalse												
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		16				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wbk.Calc ID
	1	G140H/F100LP	NRSRAPID	2	4	false	true	NONE	16	64	2061.46	
	2	G395H/F290LP	NRSRAPID	4	1	false	true	NONE	16	16	858.942	

Proposal 8063 - Observation 11 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

Special Requirements

Aperture PA Range 306.97164917 to 326.97164917 Degrees (V3 168.0 to 188.0)

Group Observations 11, 12, Non-interruptible

Aperture PA Offset 12 from 11 by 10 to 22 Degrees (Same offsets in V3)

Proposal 8063 - Observation 12 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

Observation	Proposal 8063, Observation 12: kap And b roll 2											Mon Jul 14 23:00:13 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(kap And b roll 2 (Obs 12)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
	(Visit 12:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(6)	-kap-And	RA: 23 40 24.5076 (355.1021150d) Dec: +44 20 2.16 (44.33393d) Equinox: J2000				Proper Motion RA: 80.73 mas/yr Proper Motion Dec: -18.69999998689309 mas/yr Parallax: 0.019370000000000002" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star											
	Description=[B stars]											
Template	Extended=NO											
	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		16				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G140H/F100LP	NRSRAPID	2	4	false	true	NONE	16	64	2061.46	
	2	G395H/F290LP	NRSRAPID	4	1	false	true	NONE	16	16	858.942	

Proposal 8063 - Observation 12 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

Special Requirements

Aperture PA Range 306.97164917 to 326.97164917 Degrees (V3 168.0 to 188.0)

Group Observations 11, 12, Non-interruptible

Aperture PA Offset 12 from 11 by 10 to 22 Degrees (Same offsets in V3)

Proposal 8063 - Observation 13 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

Observation	Proposal 8063, Observation 13: HR 2562 b roll 1												Mon Jul 14 23:00:13 GMT 2025
	Diagnostic Status: Warning												
Observing Template: NIRSpec IFU Spectroscopy													
Diagnostics	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(HR 2562 b roll 1 (Obs 13)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(7)	HR-2562	RA: 06 50 1.0151 (102.5042296d) Dec: -60 14 56.92 (-60.24914d) Equinox: J2000				Proper Motion RA: 4.83 mas/yr Proper Motion Dec: 108.527 mas/yr Parallax: 0.0294738" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[F stars]												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points			
	1	CYCLING		SMALL		1		16					
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wbkk.Calc ID	
	1	G140H/F100LP	NRSIRS2RAPID	8	3	false	true	NONE	16	48	6302.4		
	2	G395H/F290LP	NRSIRS2RAPID	8	1	false	true	NONE	16	16	2100.8		
Special Requirements	Aperture PA Range 45.97164917 to 65.97164917 Degrees (V3 267.0 to 287.0)												
	Group Observations 13, 14, Non-interruptible Aperture PA Offset 14 from 13 by 8 to 20 Degrees (Same offsets in V3)												

Proposal 8063 - Observation 14 - Beyond C/O and metallicity: Can refractory abundances reveal the origin of the largest planets?

Observation	Proposal 8063, Observation 14: HR 2562 b roll 2												Mon Jul 14 23:00:13 GMT 2025
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (HR 2562 b roll 2 (Obs 14)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(7)	HR-2562	RA: 06 50 1.0151 (102.5042296d) Dec: -60 14 56.92 (-60.24914d) Equinox: J2000				Proper Motion RA: 4.83 mas/yr Proper Motion Dec: 108.527 mas/yr Parallax: 0.0294738" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[F stars]												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points			
	1	CYCLING		SMALL		1		16					
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wbk.Calc ID	
	1	G140H/F100LP	NRSIRS2RAPID	8	3	false	true	NONE	16	48	6302.4		
	2	G395H/F290LP	NRSIRS2RAPID	8	1	false	true	NONE	16	16	2100.8		
Special Requirements	Aperture PA Range 45.97164917 to 65.97164917 Degrees (V3 267.0 to 287.0) Group Observations 13, 14, Non-interruptible Aperture PA Offset 14 from 13 by 8 to 20 Degrees (Same offsets in V3)												